545/3 Inst. Sch. A
CHEMISTRY
Practical
Instructions
Oct./Nov. 2026

UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Certificate of Education

CHEMISTRY PRACTICAL INSTRUCTIONS

545/3 Inst. Sch. A

October/November, 2026

CONFIDENTIAL

This information is given only to facilitate preparation of examination.

Great care should be taken that the information given below does not reach the candidates either directly or indirectly.

INSTRUCTIONS FOR PREPARING CHEMICALS AND APPARATUS

The head teacher **must** ensure that the teacher responsible for preparing the chemicals and apparatus hands in his/her procedure together with trial results properly sealed in a separate envelope and **firmly** fastened (attached) to the candidates' scripts envelope(s).

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1. The description of the reagents and chemicals specified below does **not** necessarily correspond with the description in the examination paper. Candidates must **not** be informed of the differences.

2. Candidates are **not** allowed to use reference books (i.e., text books, booklets on quantitative analysis etc.) during the examination.

3. In addition to the fittings and substances ordinarily contained in a chemistry laboratory, each candidate will require:

   1 burette (50 cm^3).

   1 thermometer (-10 ^oC -110 ^oC).

   1 boiling tube.

   1 pipette (20 or 25 cm^3).

   1 measuring cylinder (10 or 50 cm^3).

   1 stop clock/ watch.

   2 conical flasks.

4. Candidates should have easy access to:
   - Heat Source.
   - Phenolphthalein and Methyl orange indicators.

5. The school should stock the following:
   - Distilled water.
   - Concentrated sulphuric acid (1.84 g/cm^3, 98 %).
   - Adhesive stickers or masking tape for labelling.

6. Additional chemicals and instructions for preparation will be provided by **UNEB**.

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CHEMISTRY PRACTICAL - Based on Confidential Apparatus List (UNEB Style)

Instructions: In addition to the fittings ordinarily contained in a laboratory, each candidate requires:

1. burette (50 cm^3)
2. thermometer (-10 to 110°C)
3. boiling tube
4. pipette (20 or 25 cm^3)
5. measuring cylinder (10 or 50 cm^3)
6. stop clock/watch
7. 2 conical flasks

Access to: heat source, phenolphthalein and methyl orange indicators.

PRACTICAL INSTRUCTIONS

QUESTION 1: DETERMINATION OF CONCENTRATION OF SULPHURIC ACID

You are provided with solution FA1 which contains dilute sulphuric acid, and FA2 which is sodium hydroxide solution. You are required to determine the concentration of FA1.

Procedure:

1. Pipette 20 cm^3 (or 25 cm^3) of FA2 into a conical flask, add 2–3 drops of phenolphthalein indicator.
2. Fill the burette with FA1.
3. Titrate until the pink colour just disappears.
4. Repeat the titration to obtain 3 concordant readings.
5. Record your results in the table below:

Titre Number	Volume of FA2 used (cm^3)	Burette reading: Final (cm^3)	Burette reading: Final (cm^3)	Burette reading: Initial (cm^3)	Volume of FA1 used / titre (cm^3)
1	"	"	"	"	"
2	"	"	"	"	"
3	"	"	"	"	"

Questions:

(a) Calculate the average titre of FA1 used in cm^3.

(b) Calculate the number of moles of NaOH present in the 20 cm^3 (or 25 cm^3) of FA2 used.
[Given: FA2 contains 4.0 g NaOH per litre; molar mass of NaOH = 40.0 g mol^{-1}].

(c) Using the equation below, calculate:

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

(i) the concentration of the sulphuric acid, FA1, in mol dm^{-3}.

(ii) the concentration of the sulphuric acid, FA1, in g dm^{-3}. [Molar mass of H_2SO_4 = 98.0^{-1}].

QUESTION 2: RATE OF REACTION

Procedure: with FA1 (acid) and solid FA3 (sodium thiosulphate).

1. Measure 50 cm^3 of FA3 solution into a boiling tube and record its initial temperature using the thermometer.
2. Add 10 cm^3 of FA1 from the measuring cylinder into the boiling tube.
3. Immediately start the stop clock, stir the mixture and note the time taken for the cross placed beneath the boiling tube to just disappear from view.
4. Repeat the experiment at different temperatures by gently heating the FA3 solution to approximately 30°C, 40°C, 50°C and 60°C before adding FA1. each temperature, record the temperature and the time taken for the cross to disappear.
5. Plot a graph of 1/time (s^{-1}) against temperature (°C).
6. From your graph, determine and explain the effect of temperature on the rate of reaction.

• *Candidates should present all calculations clearly and record all observations with SI units where applicable.*

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Section I:

Any information which the teacher responsible for preparing the apparatus and chemicals thinks may be useful to the scorers should be given on this sheet. The teacher must try out the items and record his/her procedure and results in the space provided below to guide the scorers about the apparatus and concentrations of the chemicals used.

NB: Teachers who **do not** submit their procedure and trial results will be held responsible for their candidates' performance.

Procedure and Results:

Section II:

The Invigilator in consultation with the teacher responsible for preparing the apparatus and chemicals should give details below of any difficulties experienced by particular candidates, giving their names and personal numbers. These should include;

(a) difficulties due to faulty apparatus,
(b) accidents due to apparatus or materials,
(c) physical handicaps of candidates,
(d) any other information.

Other cases of hardship e.g., illness, disability, etc., should be reported directly to **UNEB** in the normal way.

A plan of work benches, giving details by personal numbers of the places occupied by the candidates for each session, must be enclosed with the scripts.

DO NOT STAMP ANYWHERE ON THIS DOCUMENT

Invigilator's Name _____ Signature _____

Signature of the teacher responsible for preparing chemicals _____

Signature of the head teacher _____

Random Number _____

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